# PDF Report

**Team Name:** QuarkStack

**Project Name:** QuarkSketch

**Team Members:**

Niall Concannon (PO) G00433265

Daniel Balcersak (SM) G00435838

Marko Radojcic (Tester) G00436648

Kamil Ciemniejewski (DevOps) G00433270

**Module Name:** Project Management

**Assignment Name:** Project Inception

**Lecturer:** Joseph Corr

**Date:** 22/04/2026

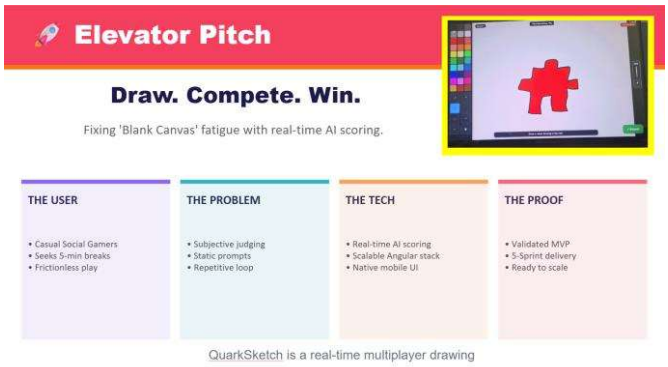
## Elevator pitch

**1. Hook / Problem:**
"Ever wanted a fast, fun way to test your drawing skills against friends—and strangers—online?"

**2. Solution / Product:**
"QuarkSketch is an online drawing showdown where players race to interpret creative prompts and outdraw their opponents.“

**3. Differentiator / Why it’s special:**
"AI generates fresh challenges, helps score each round, and players vote to crown the winner, making every match unpredictable, social, and competitive.“

**4. Call to Action / Vision:**
"Join the community and see if your drawings can beat the rest—where creativity meets competition in real time."



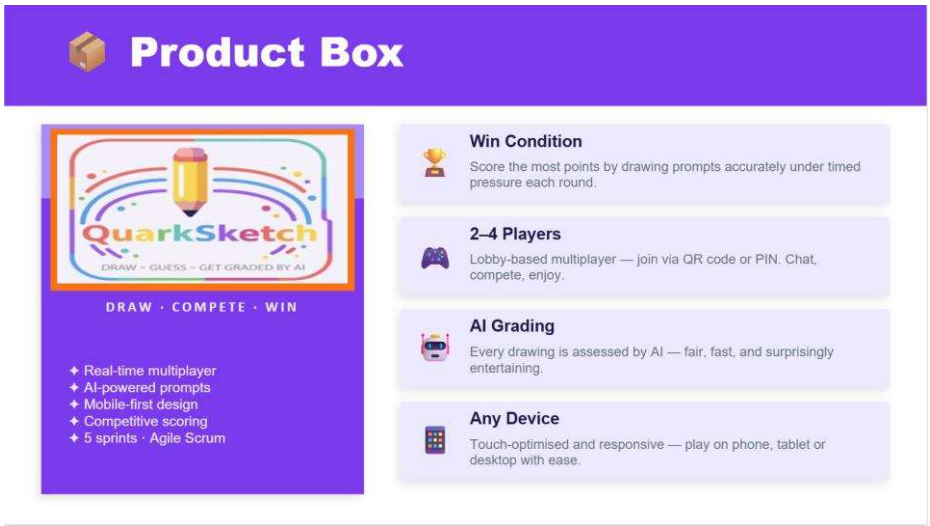

## Product box

**Name:** QuarckSketch

**Slogan:** Draw it. Guess it. Watch it go wrong.

**TOP 3 REASONS YOU'LL LOVE IT**

- Instant Fun With Friends
  Jump in and start playing immediately — no complicated setup.
- Creativity Without Pressure
  You don't need to be an artist. The worse the drawing, the funnier it gets.
- Perfect for Any Group
  Great for parties, classrooms, team-building, or online hangouts.




## Updated release plan overview

**Release Plan**

| Sprint 1 | story points | Epic | Sprint 2 | story points | Epic | Sprint 3 | story points | Epic | Sprint 4 | story points | Epic | Sprint 5 | story points | Epic |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Access main menu | 2 | F2 | Countdown timer | 2 | F1 | Warning in last 5 seconds | 1 | F1 | Fill bucket tool | 2 | F2 | Create games + choose topic (Multiplaye | 3 | F3 |
| Access canvas | 2 | F2 | Countdown before game start | 1 | F1 | Different brush sizes | 2 | F2 | Toggle SFX/Music | 1 | F6 | Chat system (text communication) | 2 | F3 |
| Select colours | 2 | F4 | Submit drawing early | 1 | F1 | Pause before round starts | 1 | F1 | AI grading system | 3 | F5 | Improve UI | 2 | F4 |
| Eraser tool | 1 | F4 | End screen with results | 2 | F1 | Prompt changes every round | 1 | F1 | Audio feedback (countdown + end soun | 2 | F6 | Improve options menu | 1 | F6 |
| Clear canvas | 1 | F4 | Show current round | 1 | F1 | Drawing history | 2 | F1 | UI click sound effects | 2 | F6 | Different game modes | 2 | F6 |
| Undo / redo | 2 | F2 | Drawing ends automatically | 1 | F1 | Line tool | 1 | F2 | | | | | | |
| | | | Background music | 2 | F6 | Shape tool | 2 | F2 | | | | | | |
| Total SPs | 10 | | | 10 | | | 10 | | | 10 | | | 10 | |

# Updated Release Plan

**S0 Project Setup**
*5–12 Mar*

- Jira board & GitHub repo
- OneNote & Microsoft Project
- Team roles & working agreement
- Release plan drafted

**S1 Core Foundations**
*12–19 Mar*

- Main menu & dark mode
- Drawing canvas built
- Eraser, undo, clear tools
- Touchscreen support

**S2 Gameplay Loop**
*19–26 Mar*

- Countdown timer system
- Auto round-end logic
- Submit button & results screen
- Background music added

**S3 Polish & Prompts**
*26 Mar–2 Apr*

- Prompt display & rotation
- Score system
- 5s warning pulse
- Brush size & drawing history

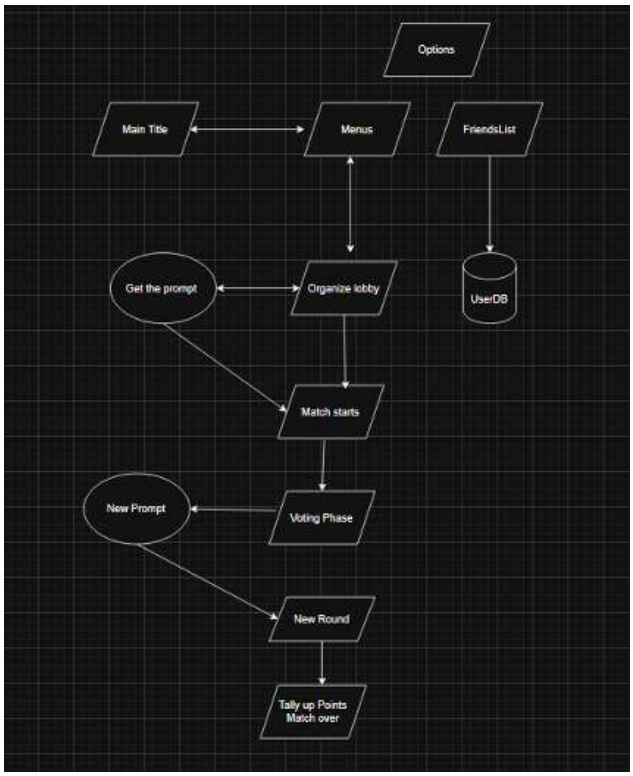
**S4 Audio & UI**
*Easter Break*

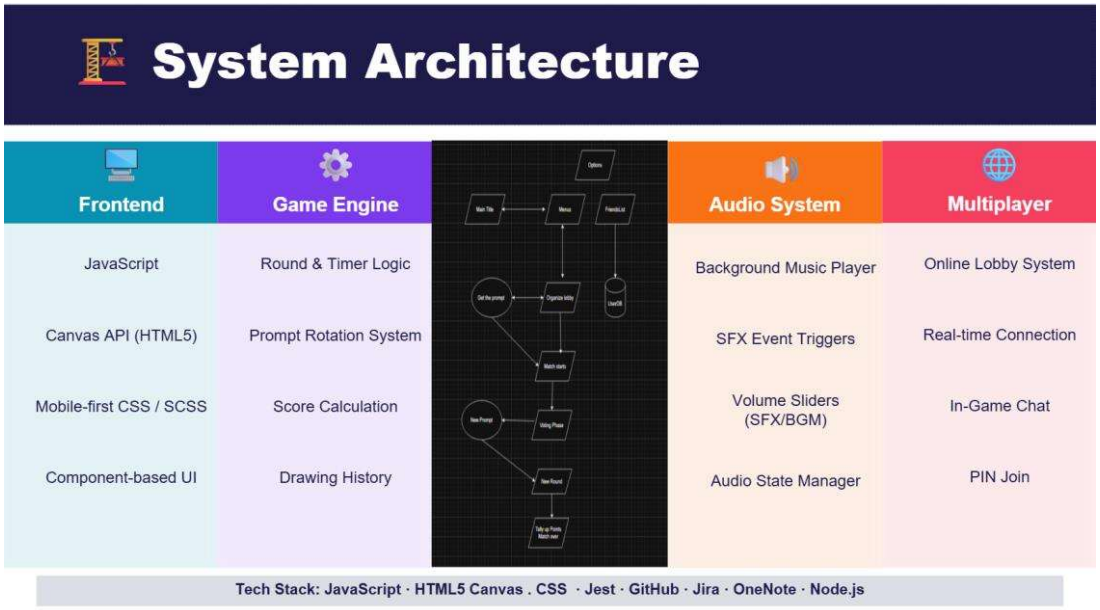
- Audio sliders SFX/music
- Mobile UI responsiveness
- Autotesting suite
- UI polish & bug fixes

**S5 Final Sprint ✓**
*14 Apr+*

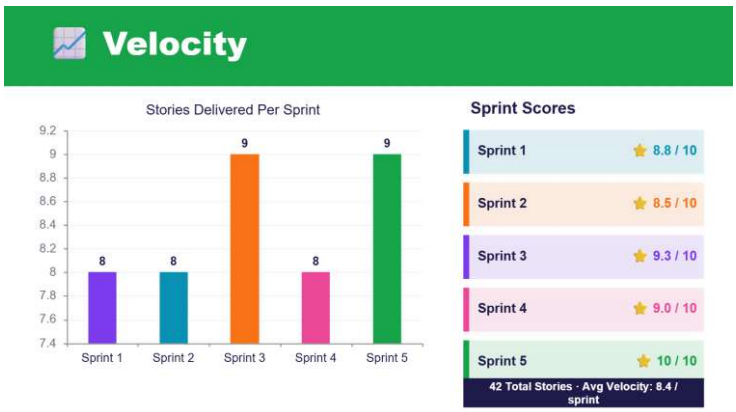
- Multiplayer & online setup
- In-game chat
- Logo redesign
- PowerPoint & documentation

## System architecture






## Velocity summary



## Sprint summaries

S0 **Project setup**
5–12 Mar 2026

8.8 / 10

HIGHLIGHTS
- Jira board & GitHub repo set up
- OneNote & Microsoft Project configured
- Team roles & working agreement defined
- Release plan drafted

BLOCKER
- Online meeting reduced work quality
- Unclear Moodle & Jira requirements
- Marko's Jira needed fixing

ACTION ITEMS
- Fix Kanban tracker in Jira
- Weekly test cases & GitHub commits

mood High overall

shoutout: Joe Corr

S1 **Core foundations**
12–19 Mar 2026

8.5 / 10

COMPLETED (8 ITEMS)
- Countdown timer
- Auto round-end logic
- Submit button & results screen
- Round indicator & pre-game countdown
- Per-round timer, lobby restart

BLOCKER
- Online Tuesday session hurt collaboration
- Bank holiday reduced in-person time
- Moodle/Jira still unclear at times

NEXT SPRINT TARGETS
- Prompt display & rotation
- Score system
- 5s warning & drawing history

mood Okay → Good

shoutouts: Joe Corr · whole team

S2 **Gameplay loop & polish**
19–26 Mar 2026

— / 10

HIGHLIGHTS
- Prompt display & rotation
- Score system implemented
- 5s warning pulse
- Brush sizes & drawing history

NOTES
- No formal retro document found
- Based on release plan deliverables

FEATURES SHIPPED
- Warning in last 5 seconds
- Different brush sizes
- Line tool & shape tool
- Drawing history view

mood Not recorded

Sprint 2 9.3 rating

**S3** Audio & UI
Easter break

9.0 / 10

| COMPLETED | BLOCKER | ACTION ITEMS |
|---|---|---|
| · Audio SFX & music controls | · Integration issues (audio + UI) | · Integration testing earlier |
| · Mobile UI responsiveness | · Test cases needed updating | · Online connection (high priority) |
| · Drawing tools refinement | · Screen-size inconsistencies | · UI consistency across screens |
| · Timer warning system | | |
| · Bug fixes & UI polish | | |

mood Good → Very Good

shoutouts: Niall · Kamil · Marko · Daniel

**S4** Final sprint
7–21 Apr 2026

10 / 10

| COMPLETED | BLOCKER | ACTION ITEMS |
|---|---|---|
| · Online multiplayer connection | · Remote Easter week slowed decisions | · Final docs proofread |
| · In-game chat feature | · Last-minute UI bugs on return | · Rehearse presentation |
| · Logo redesigned | · Multi-format docs took longer | · Per-member final build checks |
| · PowerPoint & documentation | | |
| · Full testing & validation | | |
| · Final UI refinements | | |

mood Very Good → Excellent

shoutouts: entire team

## Engineering discipline evidence

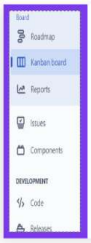
is:issue state:closed

Labels | Milestones | New issue

Open 0 | Closed 4 | Author | Labels | Projects | Milestones | Assignees | Newest

- NEED TO FIX UI AND PAINT SELECTION TOOL (HIGH PRIORITY)
  #9 · by BALCER1 was closed 17 hours ago
- need to implement multiplayer without server and instead using a selfhost and codes
  #4 · by BALCER1 was closed 2 days ago
- need to add better paint tool with larger variety
  #3 · by BALCER1 was closed 17 hours ago
- Need to fix auto testing
  #2 · by BALCER1 was closed 17 hours ago

## 🔧 Engineering Discipline Evidence



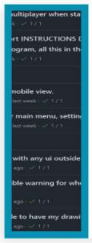

### Jira Kanban Board

✓ Sprint backlog maintained each sprint

✓ Stories written in Agile format

✓ Story points tracked & velocity measured

✓ Kanban columns: To Do → In Progress → Done



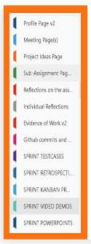
### GitHub Repository

✓ All code version controlled via Git

✓ Branch-per-feature workflow

✓ Commits linked to Jira stories

✓ github.com/Niall-Concannon/Project-Management-Group-SA3



### OneNote Documentation

✓ Sprint meeting minutes recorded

✓ Individual weekly reflections

✓ Retrospective notes & action items

✓ Evidence of work & daily standups

### Agile Process

✓ 5 full sprints completed

✓ Weekly retrospectives with scores

✓ Burndown chart tracked by Kamil

✓ Sprint planning & review sessions held

## Automated testing & acceptance test evidence

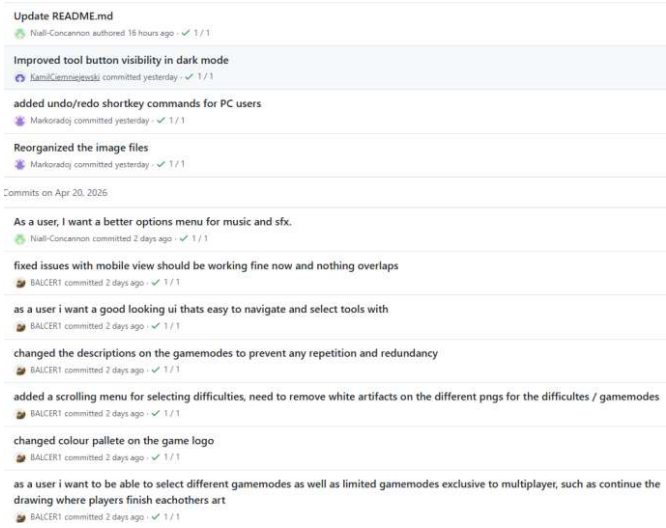
```
describe("el()", () => {
  test("creates an element with the correct tag", () => {
    const div = el("div");
    expect(div.tagName).toBe("DIV");
  });

  test("sets className via the 'class' attribute key", () => {
    const span = el("span", { class: "my-class" });
    expect(span.className).toBe("my-class");
  });

  test("sets arbitrary attributes", () => {
    const img = el("img", { src: "logo.png", alt: "Logo" });
    expect(img.getAttribute("src")).toBe("logo.png");
    expect(img.getAttribute("alt")).toBe("Logo");
  });

  test("attaches event listeners via 'on' prefix keys", () => {
    const handler = jest.fn();
    const btn = el("button", { onclick: handler });
    btn.click();
    expect(handler).toHaveBeenCalledTimes(1);
  });

  test("appends string children as text nodes", () => {
    const p = el("p", {}, "Hello World");
    expect(p.textContent).toBe("Hello World");
  });
```

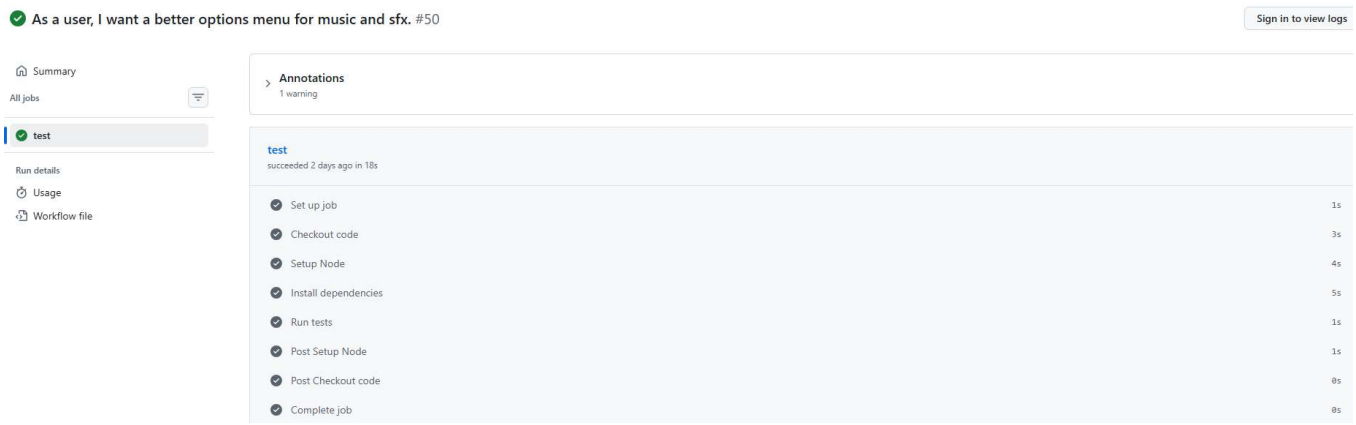
```
//result screen
const FAKE_DATA = "data:image/png;base64.abc";
const FAKE_PROMPT = { text : "Draw a Monkey Cat", subject: "animals", action: "A monkey cat

describe("resultsScreen()", () => {
test("renderes a div result screen", () => {
  expect(resultsScreen(FAKE_DATA,1, FAKE_PROMPT).classList).toContain("results-screen");
});

test("Shows 'Round Over!' heading", () => {
  expect(resultsScreen(FAKE_DATA, 1, FAKE_PROMPT).textContent).toContain("Round Over!");
});

test("Shows the correct round number", () => {
  expect(resultsScreen(FAKE_DATA, 3, FAKE_PROMPT).textContent).toContain("Round 3");
});

test("Shows the prompt text on results screen", () => {
  expect(resultsScreen(FAKE_DATA, 1, FAKE_PROMPT).textContent).toContain(FAKE_PROMPT.text);
});

test("Exit button returns to the menu" , () => {
  show(resultsScreen(FAKE_DATA,1,FAKE_PROMPT));
  document.querySelector(".btn-settings.results-btn").click();
  expect(document.querySelector(".results-screen")).toBeNull();
  expect(document.querySelector(".screen")).not.toBeNull();

});
```

```
//history helper
describe("loadHistoryEntries()",  () => {
test("returns an empty array when the local storage is empty", () => {
  expect(loadHistoryEntries()).toEqual([]);
});

const entries = loadHistoryEntries();

describe("saveHistoryEntry()", () => {
test("saves an entry on top of the list", () => {
 saveHistoryEntry({promptText: "first"});
 saveHistoryEntry({promptText: "second"});
expect(loadHistoryEntries()[0].promptText).toBe("second");
});
});

test("trims list to 40 entries max", () => {
  for (let i = 0; i < 45; i++) saveHistoryEntry({promptText: 'entry ${i}'});

expect(loadHistoryEntries().length).toBe(40);
});
});

describe("clearHistoryEntries()", () => {
  test("removes all saved data", () => {
    saveHistoryEntry({promptText: "something"});
    clearHistoryEntries();
    expect(loadHistoryEntries()).toEqual([])
  });
});
```

```
describe("countdownTimer()", () => {
  test("renders the countdown screen", () => {
  expect(countdownTimer(() => {}).className).toContain("countdown-screen");
});



test("Does not call ondone after the full sequence",  () => {
  const onDone = jest.fn();
  countdownTimer(onDone);
  jest.advanceTimersByTime(3000);
  expect(onDone).not.toHaveBeenCalled();
});

test("Tapping the screen causes a pause state", () => {
  const s = countdownTimer(() => {});
  show(s);
  const screen = document.querySelector(".countdown-screen");
  screen.click(); //pause
  jest.advanceTimersByTime(500);
  expect(screen.querySelector(".countdown-number").textContent).toBe("3");
  screen.click(); //resume
});
});
```

## Green test results



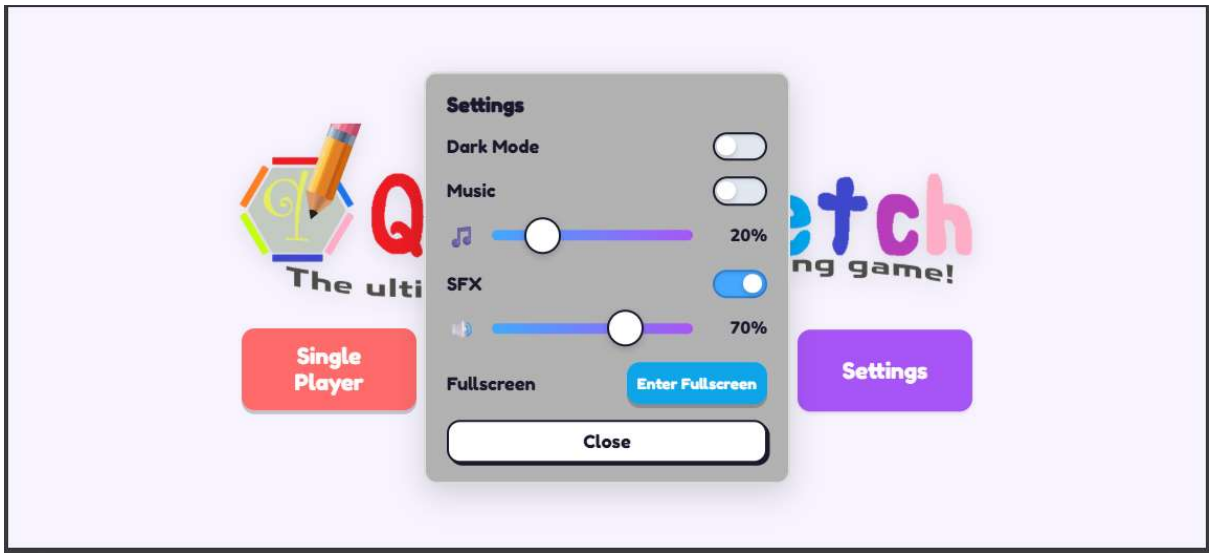



## Kit Build evidence

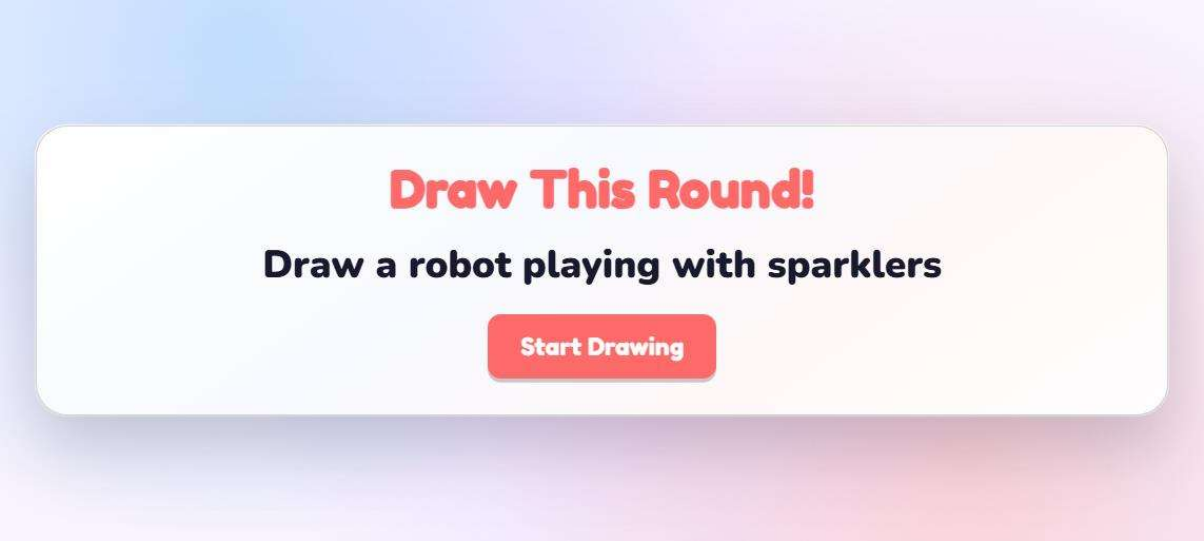
ci.yml

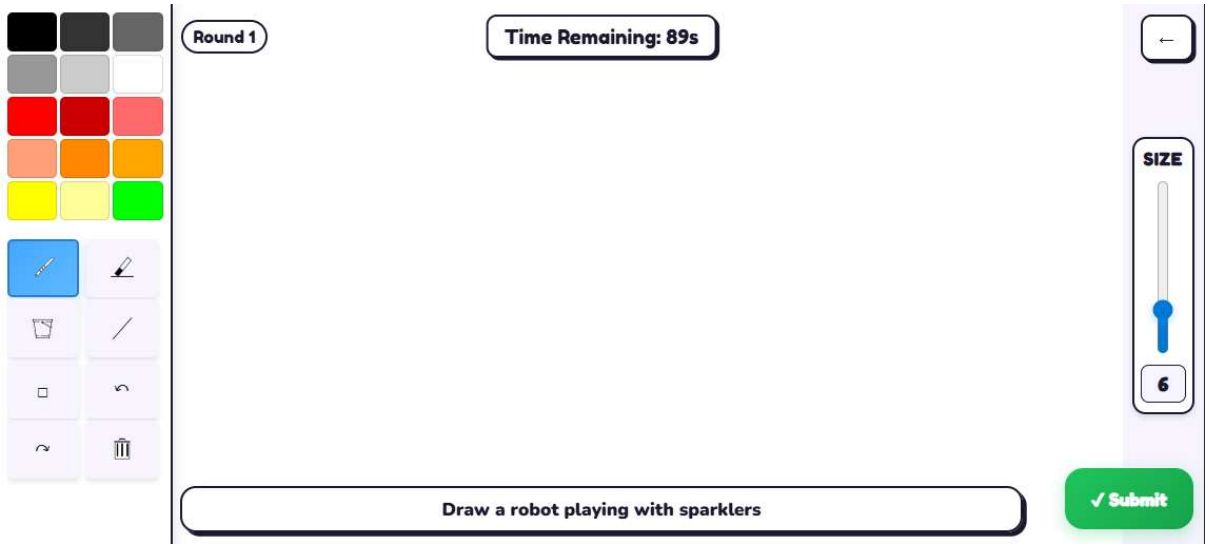
## Screenshots of final application

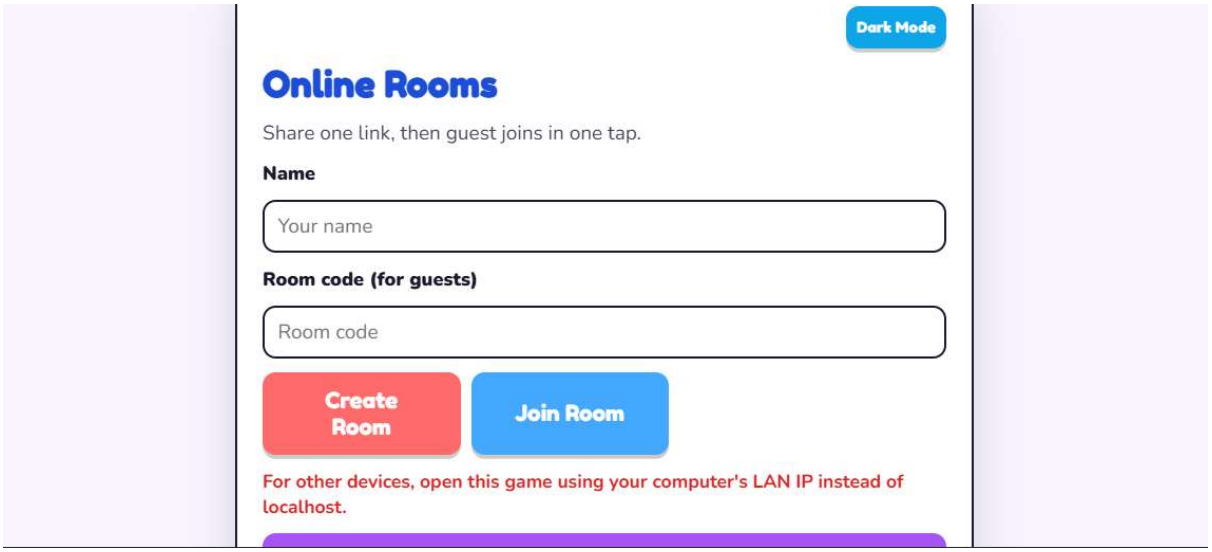
---


Settings
Dark Mode
Music
20%
SFX
70%
Fullscreen
Enter Fullscreen
Close
Single Player
Settings

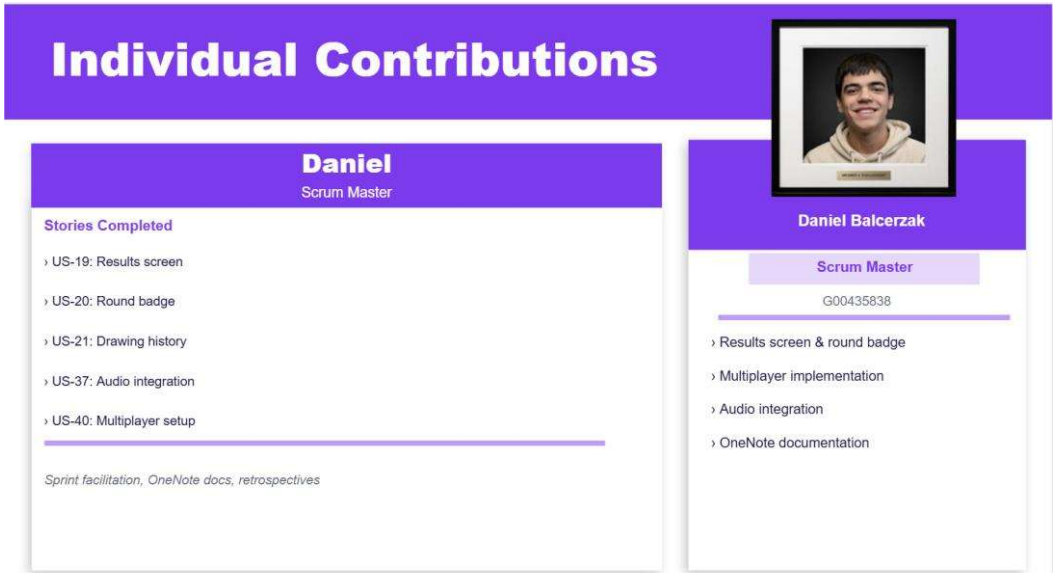
Drawing History
Clear History
Back
Draw a panther peeling a banana
Date: Apr 22, 2026, 7:25 PM
Draw time: 12s
AI: 38%
Resemblance: 48%
Color: 15%
Accuracy: 38%

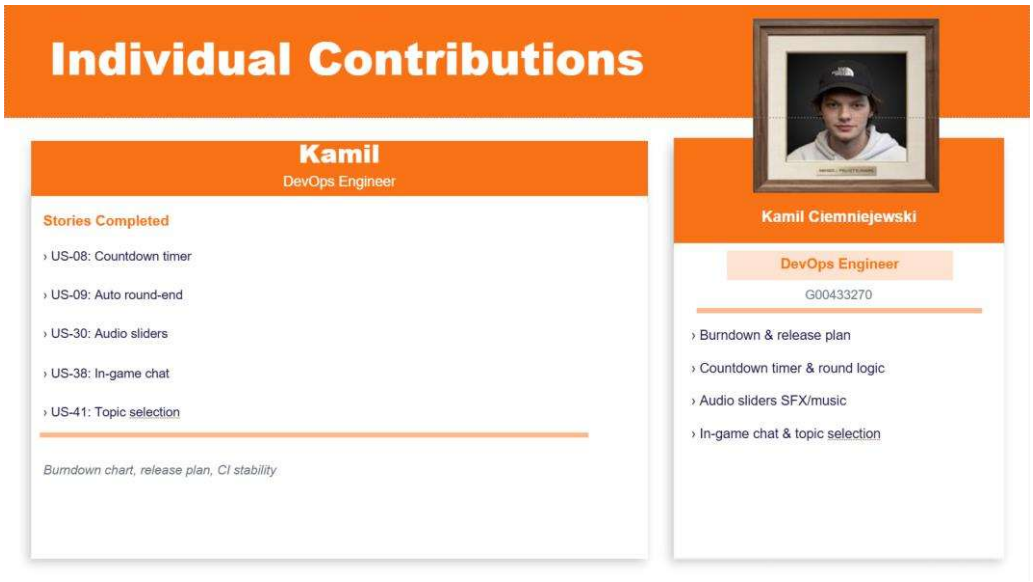

Draw This Round!
Draw a robot playing with sparklers
Start Drawing

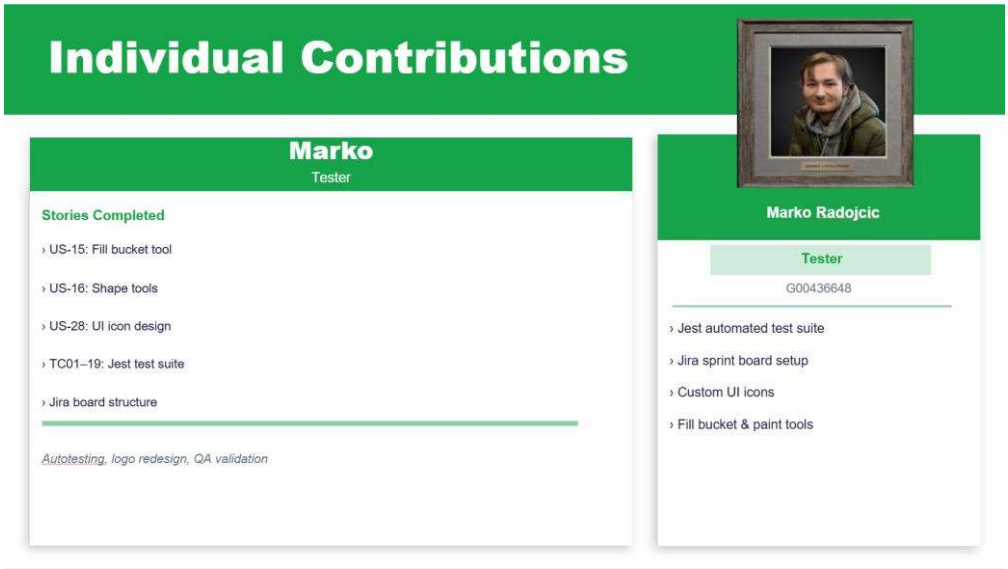

Round 1
Time Remaining: 89s
SIZE
6
Draw a robot playing with sparklers
✓ Submit


Dark Mode
Online Rooms
Share one link, then guest joins in one tap.
Name
Your name
Room code (for guests)
Room code
Create Room
Join Room
For other devices, open this game using your computer's LAN IP instead of localhost.

## Individual contributions






## Required Links (In the PDF)

**OneNote workspace -** QuarkStack

**GitHub repository -** https://github.com/Niall-Concannon/Project-Management-Group-SA3

**Jira project -** SCRUM Sprint 0 - QuarkStack - Scrum Board - Jira

**MS Project plan -** https://atlantictu-my.sharepoint.com/:u:/r/personal/g00433265_atu_ie/Documents/25-26/Project%20Management/Microsoft%20Project.mpp?csf=1&web=1&e=9Q8yOQ

**Presentation slides -** QuarkStack_Presentation_DONE 1.pptx

**Recorded demo/screencast of Final App -** Video Demo QuarkSketch.mp4

**Final kit -** ci.yml